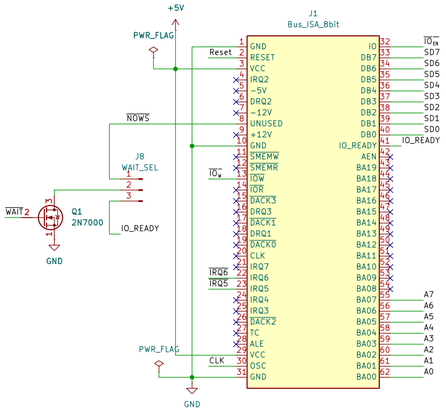
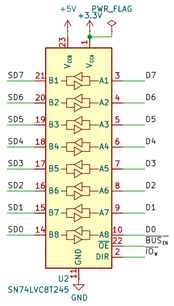


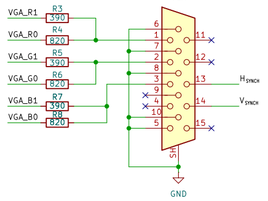


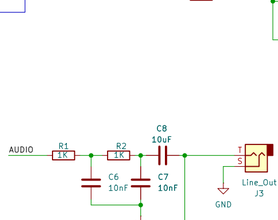

```
ESP32-S3:
// ------------------6502 bus
// All bus pins must be < 32 so one read of GPIO_IN_REG
captures everything.
// D0..D7 must be consecutive GPIOs starting at PIN_D0.
#define PIN_D0     4    // D0..D7 = GPIO4..GPIO11 (via
74LVC8T245)
#define PIN_A0     12   // register select bit 0  (from GAL,
open drain + 1k pull-up)
#define PIN_A1     13   // register select bit 1  (input)
#define PIN_RW     14   // R/W from CPU, 1 = read  (input)
#define PIN_NCS    15   // /CS for $7F18-$7F1B, decoded
by the GAL, active low
#define PIN_PHI2   16   // PHI2 clock (input)
#define PIN_ACK    17   // to GAL ACK: 0 = hold RDY low
while selected
#define PIN_IRQ    18   // to GAL IRQREQ: 1 = pull /IRQ low

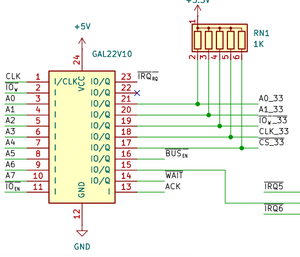
// ------------------VGA
// 2-bit-per-channel resistor DAC like FabGL (64 colours).
// Pixel byte format: 0b00RRGGBB.
#define PIN_VGA_B0   46
#define PIN_VGA_B1   45
#define PIN_VGA_G0   48
#define PIN_VGA_G1   47
#define PIN_VGA_R0   42
#define PIN_VGA_R1   41
#define PIN_VGA_HSYNC 39
#define PIN_VGA_VSYNC 40

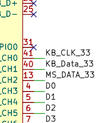
// ------------------PS/2
// Through a BSS138 4-channel bidirectional level shifter (5 V
<-> 3.3 V).
#define PIN_KBD_CLK  1
#define PIN_KBD_DAT  2
#define PIN_MOUSE_CLK 21
#define PIN_MOUSE_DAT 3

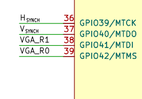
// ------------------Audio
// Sigma-delta output -> RC low-pass filter -> amplifier / line
out.
#define PIN_AUDIO   38
```

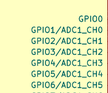


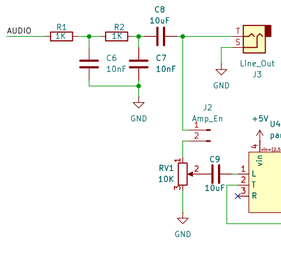


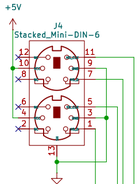













```
GAL22V10:
/* ------------------------------- Inputs */
PIN 1  = PHI2 ; /* AKA CLK  */
PIN 2  = RW ;  /* AKA IOw  */
PIN 3  = A0 ;
PIN 4  = A1 ;
PIN 5  = A2 ;
PIN 6  = A3 ;
PIN 7  = A4 ;
PIN 8  = A5 ;
PIN 9  = A6 ;
PIN 10 = A7 ;
PIN 11 = nIO_EN ; /*  */
PIN 13 = ACK ;    /* ESP32 GPIO17: 0 = hold RDY low while selected AKA DMA */
PIN 23 = IRQREQ ;  /* ESP32 GPIO18: 1 = pull /IRQ low          */

/* ------------------------------------------------- outputs */
PIN 14 = RDY ;    /* to CPU RDY    (open drain, 5 V pull-up on SBC) */
PIN 15 = nIRQ ;   /* to CPU /IRQ   (open drain, 5 V pull-up on SBC) */
PIN 16 = nBUSEN ;  /* to 74LVC8T245 /OE (push-pull)          */
PIN 17 = nCS_33 ;  /* ESP32 GPIO15  (open drain, 1 k to 3.3 V)  */
PIN 18 = PHI2_33 ;  /* ESP32 GPIO16  AKA CLK_33              */
PIN 19 = RW_33 ;   /* ESP32 GPIO14  AKA IOw_33              */
PIN 20 = A1_33 ;   /* ESP32 GPIO13                          */
PIN 21 = A0_33 ;   /* ESP32 GPIO12                          */
```

H1 MountingHole

H2 MountingHole